

Education

The University of Cambridge

October 2022 - present

PhD in Engineering, Bioengineering

Cambridge, UK

- Thesis: improving and maintaining the *in vitro* maturity of cardiomyocytes; submission expected September 2026.
- Co-supervised by the Biointerface Group (Department of Engineering) and the Li Lab (Department of Medicine).

Trinity College Dublin

September 2018 - May 2022

Baccalaureus in Arte Ingeniaria (B.A.I), Biomedical Engineering

Dublin, IE

- First Class Honours
- Thesis: 'Metabiomaterial: Multi-layered 3D immunomodulating patch'

Publications

Journal Articles

- **J.F. Murphy** *et al.* A perspective towards building scalable, integrated, multicellular *in vitro* airway models with inverted airway-on-a-tube. *Bio-Design and Manufacturing*. doi:10.1631/bdm.2500615 (2026)
- **J.F. Murphy** and X. Li. The roles of the microtubule cytoskeletal network in cardiac mechanobiology. *Journal of Cell Science*. doi:10.1242/jcs.264308 (2026)
- A. Doryab *et al.* Hydrogel-based airway-on-tube with perfusable endothelial lumen and outward epithelialization. *Advanced Materials*. doi:10.1002/adma.202523588 (2026)
- **J.F. Murphy** *et al.* Biofabrication and biomanufacturing in Ireland and the UK. *Bio-Design and Manufacturing*. doi:10.1007/s42242-024-00316-z (2024)
- Y. Pan *et al.* Fabrication, sustainability, and key performance indicators of bioelectronics via fiber building blocks. *Cell Reports Physical Science*. doi:10.1016/j.xcrp.2024.101930 (2024)
- **J.F. Murphy** *et al.* Rianú: Multi-tissue tracking software for increased throughput of engineered cardiac tissue screening. *Computer Methods and Programs in Biomedicine Update*. doi:10.1016/j.cmpbup.2023.100107 (2023)
- **J.F. Murphy** *et al.* Adult human cardiac stem cell supplementation effectively increases contractile function and maturation in human engineered cardiac tissues. *Stem Cell Research and Therapy*. doi:10.1186/s13287-019-1486-4 (2019)
- J. Mayourian *et al.* Exosomal microRNA-21-5p mediates mesenchymal stem cell paracrine effects on human cardiac tissue contractility. *Circulation Research*. doi:10.1161/CIRCRESAHA.118.312420 (2018)

Preprints

- M. Suku *et al.* Synergistic generation of cardiac resident-like macrophages and cardiomyocyte maturation in tissue engineered platforms. *bioRxiv*. doi:10.1101/2024.12.04.626684 (2024)

Book Chapters

- I.C. Turnbull *et al.* Cardiac tissue engineering models of inherited and acquired cardiomyopathies. *Methods in Molecular Biology*. doi:10.1007/978-1-4939-8597-5_11 (2018)

Conference Abstracts

- **J.F. Murphy** *et al.* *Bio-Design and Manufacturing Conference (BDMC)*. (2025)
- **J.F. Murphy** *et al.* *BioMedEng24*. (2024)
- **J.F. Murphy** *et al.* *TERMIS World Congress*. (2024)
- **J.F. Murphy** *et al.* *Biomedical Engineering Society Annual Meeting*. (2019)
- S.I. Salazar *et al.* *New York Academy of Sciences*. (2019)
- **J.F. Murphy** *et al.* *American Heart Association Scientific Sessions*. (2018)
- I.C. Turnbull *et al.* *International Society for Stem Cell Research*. (2018)

Research Experience

Biointerface Group, University of Cambridge

October 2022 - present

PhD Researcher

Cambridge, UK

- Designed, simulated, and fabricated a bulk acoustic wave device for label-free cell patterning, deriving transducer resonance analytically from piezoelectric compliance and stiffness matrices and validating the prediction against impedance analysis and 2D/3D simulation.

- Demonstrated acoustic alignment of primary murine cardiomyocytes and *ex vivo* cardiac slices, exploiting the acoustic torque on rod-shaped cells to orient them relative to pressure nodes.
- Engineered a synthetic adhesion protein targeting the cardiomyocyte intercalated disc, combining AlphaFold 3 structure and interaction prediction with viral expression and confocal localisation.
- Built electrical stimulation and microgroove culture platforms and developed ECG-based functional readouts for *ex vivo* cardiac tissue.

Monaghan Lab, Trinity Centre for Bioengineering

September 2018 - May 2022

Research Assistant

Dublin, IE

- Characterised a novel biomaterial's mechanical and physicochemical properties through tensile testing, ELISA assays, FTIR analysis, mass spectroscopy, flow cytometry, and contact angle assessment.
- Provided expertise to PhD students on 3D design and printing, induced-pluripotent stem cell culture and differentiation, and engineered cardiac tissue fabrication techniques.
- Key exchange point of international collaboration between the Costa Lab (NYC) and the Monaghan Lab (Dublin). Focused on adapting an engineered cardiac tissue bioreactor for use with a novel pacing system.
- Performed histological staining and analysis of tissue explants with polarised light microscopy to characterise fibrotic encapsulation around next-generation silicone implants.
- Carried out in-depth image analysis using threshold segmentation and region-of-interest normalisation.
- Developed a setup to determine propagation of electric pulses across biomaterial bridge between fresh muscle tissues *ex vivo*.

Turnbull Lab, Icahn School of Medicine at Mount Sinai

June 2021 - September 2021

Research Assistant

New York City, USA

- Developed Rianú, a web application capable of tracking and analysing multiple cardiac tissues simultaneously.
- Modified the existing tissue recording setup to record multiple tissues in parallel.

Costa Lab, Icahn School of Medicine at Mount Sinai

March 2017 - September 2020

Research Assistant

New York City, USA

- Maintained induced-pluripotent stem cells (iPSCs), mesenchymal stem cells, and cardiac stem cells in culture.
- Differentiated iPSCs into cardiomyocytes and fabricated 3D human engineered cardiac tissues. Used engineered tissue as a testing platform for various drug- and cell-based therapies for cardiac regeneration.
- Used existing LabVIEW and MATLAB software to collect and analyse data on cardiac function.
- Designed and printed 3D accessories to help with the data collection process.

Technical Skills

Cell and Tissue Culture

- Induced-pluripotent, mesenchymal, and cardiac stem cell maintenance; cardiomyocyte differentiation; fabrication of 3D human engineered cardiac tissues; primary murine cardiomyocyte isolation; murine and rat cardiac slice fabrication and *ex vivo* culture.

Characterisation

- Tensile and mechanical testing, ELISA, FTIR, mass spectrometry, flow cytometry, contact angle measurement, histological staining, impedance analysis, ECG.

Microscopy and Image Analysis

- Confocal, polarised light, brightfield, and video microscopy; threshold segmentation, region-of-interest normalisation, and orientation/coherency analysis in Fiji.

Simulation and Software

- COMSOL Multiphysics, MATLAB, LabVIEW, Fiji/ImageJ, AlphaFold 3 (PyMOL, ChimeraX), L^AT_EX.

CAD and Fabrication

- SolidWorks, Autodesk Fusion 360, Inventor and Revit, OpenSCAD; 3D design and printing, laser cutting, piezoelectric transducer assembly, GelMA hydrogel casting.

Teaching Experience

Department of Engineering, University of Cambridge

October 2023 - January 2026

Demonstrator, Part IB Structures and Materials — Materials

Cambridge, UK

Characterisation Laboratory (M1)

- Demonstrated materials characterisation techniques to Part IB undergraduates, spanning commercial instrumentation (Instron testing machine) and low-cost methods including beam vibration, plate vibration via Chladni patterns, beam bending, and video microscopy.
- Assessed characterisation mini-projects, marking written reports and giving feedback on group presentations.

- Supported first-year undergraduates through their SolidWorks course, staffing help desks to help part modelling and simulations.
- Marked assessed CAD tasks at the term's marking sessions and gave individual feedback on modelling approach.

Open Source Projects

Rianú: Multi-tissue tracking software for increased throughput of engineered cardiac tissue screening

<https://rianu.mrph.dev>

- Identified a bottleneck in existing engineered cardiac tissue analysis software.
- Created a web application capable of tracking and analysing multiple engineered cardiac tissues simultaneously.
- Validated this software against existing software and provided detailed documentation for its use.
- Released open-source under the BSD 3-Clause License and actively maintained. Source: gitlab.com/hect-software/rianu; archived at [doi:10.17605/OSF.IO/YWCHZ](https://doi.org/10.17605/OSF.IO/YWCHZ).

Awards and Activities

Awards

- Best Paper, Bio-Design and Manufacturing Annual Conference (2025)
- W.D. Armstrong Studentship, School of Technology, University of Cambridge (2022 - 2026)
- Honorary Robert Gardiner Memorial Scholarship, University of Cambridge (2022)
- Book Prize, Department of Engineering, Trinity College Dublin (2019, 2020, 2021)

Leadership and Service

- Organising Committee, Annual Division C Conference (Mechanics, Materials and Design), Department of Engineering, University of Cambridge (2023)
- Captain, Cambridge University Blues Men's Ice Hockey Team (2025-2026)
- President, Cambridge University Ice Hockey Club (2025-2026)
- Sports Personality of the Year, University of Cambridge (2024)

Citizenships

- Republic of Ireland
- United States of America